
Finding What Experts Leave Unsaid

The proof of concept, what it shows, and the research program that could validate it

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Scope: the zero-participant NOW program, items N0–N3 (evidence model, evidence reconstruction, gap map), and the program that would validate them.

Companion document: the Extended Technical Report, which holds protocols, full results, implementation details and supplementary analysis. Validation Track A (physics, human expert elicitation) is frozen and is not reported as a result here.

How to read this report

This report separates five kinds of statement, and marks them where the kind is not obvious from the wording: **L** LIT. a result from published work; **O** OBS. something this proof of concept (PoC) produced, checked against its own output files, code or logs; **I** INT. our reading of what an observation means; **H** HYP. a claim we think is plausible but have not tested; **F** FUT. work not yet done. A FINDING box states a conclusion we are prepared to defend. A KEY QUESTION box states something we do not know yet. In figures, a solid outline means *implemented in the PoC*; a dashed outline means *planned, not built*. Repository paths such as `research/n3/README.md` link to that file at the audited commit named on the title page, not to the moving main branch. This Main Research Report carries the argument; the Extended Technical Report holds the protocols, full results, implementation details and supplementary analysis, and pointers such as “(Extended Technical Report, Section 17)” refer to it.

The one distinction that matters most: the PoC *predicts* where hidden expert knowledge may lie. No human has yet checked those predictions. A predicted gap is not observed expert knowledge.

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1 Summary

The result. o **OBS.** The proof of concept (PoC) built a chain from public web text to ranked questions for experts, and it runs. Its central step does not work yet. That step, the absence check (*closure* in the code), asks a judge model whether retrieved evidence already states an element that a methodology lens flagged as missing. In a control, each candidate's additional other-source sentences were swapped for those of the topically nearest candidate, while the seed's own sentences stayed in both arms. The judge's closure rate did not change: 0.889, 0.961 and 0.958 in the three ledgers, in both arms. Under this control, the judge did not tell a candidate's own cross-source evidence from the donor's, so it is not yet an informative absence check; the test does not show why. No gold standard was acquired, and no human has checked any prediction. The PoC therefore neither shows nor refutes that the gap map predicts what experts know.

The recommendation: run the closure study before any grant. f **FUT.** The cheapest decisive test needs no grant and no participants. In this Phase 0 study (V1), blind coders who are not the owner (two preferred) label each unit as closed, partly closed or open. The sample is the 107 lens firings from the two admissible ledgers (105 with a donor in both arms of the control), plus about 100 units from one new domain; the firing, not the arm, is the unit of inference. V1 reports coder agreement, corpus-level retrieval recall, and whether any judge in a panel beats the control. It costs coder hours and local compute, on the order of EUR 10³. Its gate, MS1, reads CONTINUE only if coder $\kappa \geq 0.70$, retrieval recall meets its floor (proposed 0.80), and at least one judge beats the control by a pre-registered margin. Agreement shows the label is reproducible, not that it is right. Otherwise the automated absence check stops. The V1 numbers belong on page 1 of any grant proposal (Section 10).

The problem and the idea. l **LIT.** Experts leave out the steps that have become automatic for them. In one small study, three surgeons teaching a procedure omitted on average 73% of the decision steps in a task list built by cognitive task analysis (CTA) (Sullivan et al. 2014); no general omission rate is known. Writers also leave out what is obvious to them (Gordon and Van Durme 2013). We ask AI for two narrow things: *reconstruct* what public sources state about a task, as a ledger of labeled claims tied to exact source spans; and *predict where that reconstruction is thin*, using constructs from expertise research as lenses. What survives the absence check is a *gap candidate*: a hypothesis labeled *inferred*, with a template question for an expert.

What was built and shown. o **OBS.** With zero participants and public web text only, in two domains (intermittent faults in programmable logic controllers, PLC; data protection impact assessments under the GDPR), at \$0.50–\$1.00 per run:

- an evidence model with computed labels, gates that refuse synthetic criteria, and a frozen schema (N1);
- a reconstruction pipeline with verbatim spans and cross-family verification (N2). All 909 stored evidence items re-slice exactly from their snapshots; 260 of 1,169 claims have no locatable span and 158 more have a span the verifier did not accept; all 418 are labeled synthetic. N2's registered research verdict is INCONCLUSIVE;
- a gap map with seven lenses, absence check, triage, a breadth ranking and template questions (N3). On the two admissible ledgers the lenses fired 107 times and left 25 gap candidates after merging; built-in checks expose the absence check's failure on every replay;

- an exploratory spike with a low-cost decision model, Jev, on the committed N3 artifacts: its per-record “tacit knowledge” score showed NO SIGNAL under the pre-set form test: it tracked text form.

Not shown: that any candidate is real expert knowledge; that the absence check works; that gating resists planted falsehoods; that the ranking is more than source density; that anything generalizes beyond two domains.

What is new. **I** **INT.** Every step has precedent; the closest work checks GDPR privacy policies for regulation-required content (Torre et al. 2020). What we did not find is a method that chains a provenance-typed ledger, expertise constructs as absence detectors, a knowledge type and elicitation channel per candidate, and search failures kept apart from expert questions. That chain is a proof of concept, not a validated method.

The program after V1. **F** **FUT.** Each tranche of money is released only by a gate reading CONTINUE, read by an external gate reader, not the owner. *Phase 1*, a lean methods grant (about EUR 0.3–0.5 M, 12–15 months, only on MS1 CONTINUE), freezes the configuration and screens the gap map against several published CTA task lists; its gate MS3 releases Phase 2 money but does not decide the central hypothesis H2. *Phase 2* (EUR 0.7–2.2 M, only on MS3 CONTINUE) tests H2 by a blind prospective study with a provisional smallest gain of interest $\delta = 0.05$ in area under the ROC curve (AUROC), sized after its pilot. **Team status:** one operator built the PoC with AI assistance; a host institution, principal investigator, statistician, CTA collaborator and gate reader must be named before submission (Section 12).

2 The problem

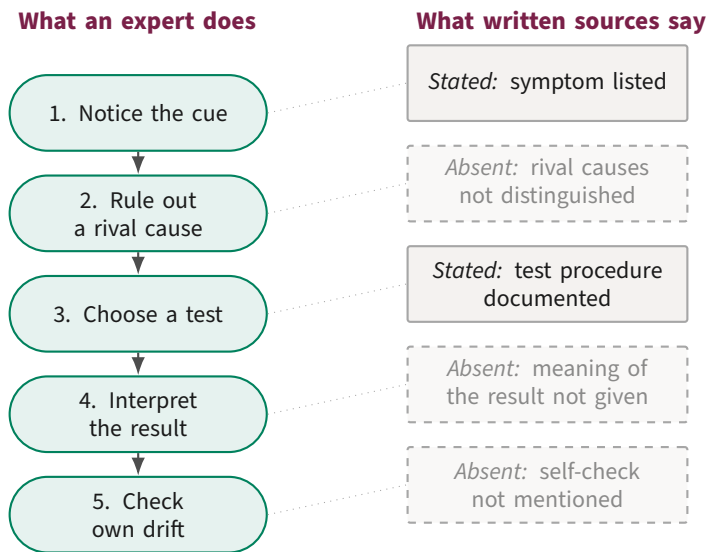
L **LIT.** Practice compiles a sequence of decisions into a fast, automatic procedure that no longer needs the knowledge it was built from (Anderson 1982). Its steps become “not available for introspection” (Clark et al. 2008, p. 584). An expert who explains a familiar task therefore omits steps, not out of reticence, but because they no longer come to mind. How much any one report leaves out is an empirical question.

2.1 What omission looks like when it is measured

L **LIT. ESTABLISHED LITERATURE**

Three surgeons, videotaped teaching a cricothyrotomy, were compared against a 46-step task list built by CTA from three other surgeons. On average they omitted 71% of clinical-knowledge steps (10 of 14), 51% of action steps (14 of 27) and 73% of decision steps (3.6 of 5), and described *how* to perform only 13% of action steps. Directed CTA prompting raised the share of steps each surgeon described from 44% (20 of 46) to 66% (31 of 46) (Sullivan et al. 2014).

L **LIT.** Two other studies give numbers. Six expert programmers explained their troubleshooting under several elicitation methods (Chao and Salvendy 1994). As reported by Clark et al. (2008, p. 585) (we did not check the original tables), no single expert reported more than 41% of diagnostic actions, 53% of debugging actions or 29% of interpretations; pooling all six raised coverage to 87%, 88% and 62%. Critical Decision Method (CDM) interviews with neonatal intensive-care nurses



Illustrative example: a hypothesized pattern, not measured in any study.

LITERATURE (other domains). Sullivan et al. 2014 (3 surgeons, 1 procedure): 73% of decision steps and 71% of clinical-knowledge steps omitted.

Chao & Salvendy 1994, as reported by Clark et al. 2008 (6 programmers): no single expert reported more than 41% of diagnostic actions.

Reporting bias (commonsense facts, not expert text): how often people write about something distorts how often it occurs.

○ Expert activity (trace) □ Appears in written sources ▤ Absent from written sources

Figure 1: What part of expert knowledge reaches text, and what does not. The task trace is illustrative, not measured. Automaticity and measured omission in three small studies show that experts' own reports leave out many decisions, cues and interpretations; reporting bias is shown for commonsense facts. That written expert sources omit the same kinds of knowledge is our hypothesis, not a finding of these studies.

produced a sepsis-assessment guide with information “not available in the current literature” (Crandall and Getchell-Reiter 1993).

I INT. These studies do not license a single omission rate. Each has 3 to 17 experts in one domain, and their gold standards are themselves built by CTA. The figure often quoted outside this literature, that experts are unaware of about 70% of their own decisions, is not a new measurement; Clark et al. (2008, p. 591) state it citing earlier reviews. We treat the direction as established: experts omit a large, variable share of decisions, cues and interpretations.

2.2 Writers omit the obvious too

L LIT. How often people write about actions, outcomes and properties systematically distorts how often they occur (Gordon and Van Durme 2013). Text-only language models inherit the distortion, for example in the colors they predict for 521 objects (Paik et al. 2021). Both studies concern commonsense facts, not expert documentation. **H HYP.** We transfer the idea: omission in speech and reporting bias in text may point at the same target in expert sources, namely cues, decisions, interpretations, checks and the boundaries of hedged rules. This transfer is untested. AI can make the boundary between what is written and what is not sharper and more auditable; it

cannot tell on its own whether a silence hides expert knowledge, an unrecorded fact, or nothing. That needs a human answer (Figure 1).

2.3 One education and one organizational example

L LIT. *Education.* Experts sort physics problems by the principle that solves them; novices sort them by surface features (Chi et al. 1981). For the expert, choosing the principle happens as recognition, not as a step, so it goes unsaid. Experts are also unreliable judges of what learners find hard: 25 physics teaching assistants and 30 instructors predicted students’ most common wrong answers at 65% and 68% of the maximum score, against 40% for random guessing, with no difference between the groups (Maries and Singh 2016). Hence expert or AI guesses about learner difficulty are never treated as evidence here.

L LIT. *Organizations.* Photocopier technicians diagnose through shared stories of how a machine actually fails; those stories, not the service manual, keep the group’s knowledge current (Brown and Duguid 1991; Orr 1996). **O OBS.** On public PLC troubleshooting text, the gap map produced a record of the same shape from the span “a common mistake... is panicking and abandoning systematic approaches”: the sources name the failure mode, and the absence check found at most a partial statement of how a troubleshooter notices the drift into it.¹ This shows only that a lens can flag text of this *shape*.

L LIT. *Why it matters.* In a study of best-practice transfer inside firms, the major barriers were knowledge-related (the recipient’s lack of absorptive capacity, causal ambiguity, and an arduous relationship between source and recipient) rather than mainly motivational (Szulanski 1996). Many popular open-source projects depend on one or two developers (Avelino et al. 2016), and turnover exposes projects to knowledge losses far above the expected loss (Rigby et al. 2016).

I INT. Training, onboarding and succession can suffer when such knowledge is not made explicit before its holder leaves.

3 Question and hypotheses

? KEY QUESTION

RQ-B. Can AI reconstruct what is written about a domain, and predict where its own reconstruction is thin, so that the location of the human knowledge residual is identified before any human is asked?

H1 – Reconstruction with provenance is feasible and checkable. A system extracts atomic claims from public sources, ties every evidence item to the exact span it came from, and labels each claim’s evidence status. H1 has an adversarial part: gating lets fewer planted falsehoods and false answers through than the ungated model.

H2 – Lenses plus an absence check predict where experts add unwritten knowledge, by a margin that matters. Constructs from CTA and related methods, applied as lenses and followed by a check of whether the expected element is stated, rank areas by where a fixed elicitation protocol brings out validated knowledge the evidence lacks. Their area-level AUROC exceeds that of the strongest baseline without this machinery, such as text density or a plain language-model

¹ research/n3/plc/gapmap.json, the GUARD-lens record at rank 9.

guess, by at least δ , a smallest effect size of interest (SESOI). A real but smaller gain does not satisfy H2. *H2 is the central hypothesis.*

F FUT. FUTURE WORK

The falsifier. Δ AUROC is the gap map’s area-level AUROC for ranking areas where the protocol elicits residual knowledge above the others, minus that of the strongest pre-registered baseline. H2 is falsified if the upper bound of a 90% confidence interval on Δ AUROC falls below δ , fixed before any gold is acquired. H2 survives if the lower bound exceeds 0 and the point estimate reaches δ ; that rejects “no gain” but does not establish a gain of at least δ , which only a lower bound above δ would. $\delta = 0.05$ is a provisional program threshold (Section 10). The blind prospective study (V5) reads this rule. It has not run.

H3 — Prioritized questions reduce expert time. Questions chosen from the gap map recover more validated, previously unwritten items per expert hour than an untargeted interview of equal length, by a ratio of at least 1.25 (H3’s SESOI, fixed before data); V6 reads it with the same rule form as H2.

I INT. H2 means something only if H1 holds, and H3 only if H2’s ranking beats chance. **H** HYP. A separate hypothesis sits outside this chain. **H-OSS:** the gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave. It concerns organizational artifacts, not the human residual, so it can neither support nor refute H2.

Table 1: What the zero-participant PoC could test for each hypothesis, and its status. No cell reports a result against human knowledge.

Hyp.	What the PoC could test	Status after the PoC
H1	Whether a ledger with verbatim provenance and cross-model verification runs end to end on real sources	O OBS. Engineering part: yes for the pre-fix pipeline on PLC v1 and GDPR v1; the fixed pipeline has not completed a live run. Adversarial part: N2’s registered verdict is INCONCLUSIVE; the gated arms were never scored (Section 6.1).
H2	Whether lenses fire on real text and the absence check separates “unstated” from “stated elsewhere”	O OBS. Mechanism only. Lenses fire on three ledgers; the absence check ties its mismatched-evidence control. No Δ AUROC computed. Next test: V1, before any grant.
H3	Whether ranked questions can be produced at all	F FUT. Not tested. Template questions exist; no expert has answered one.

4 From existing methods to the approach

Several research traditions each solve part of the problem. None answers the question this project asks: *in this domain, where does the written record stop, and what kind of knowledge is likely to lie beyond it?* This section first states the approach, then what each tradition contributes to it, then the closest prior work.

4.1 The central model

Figure 2 shows the approach. Steps 1–6 are built in the PoC; steps 7–9 are not.

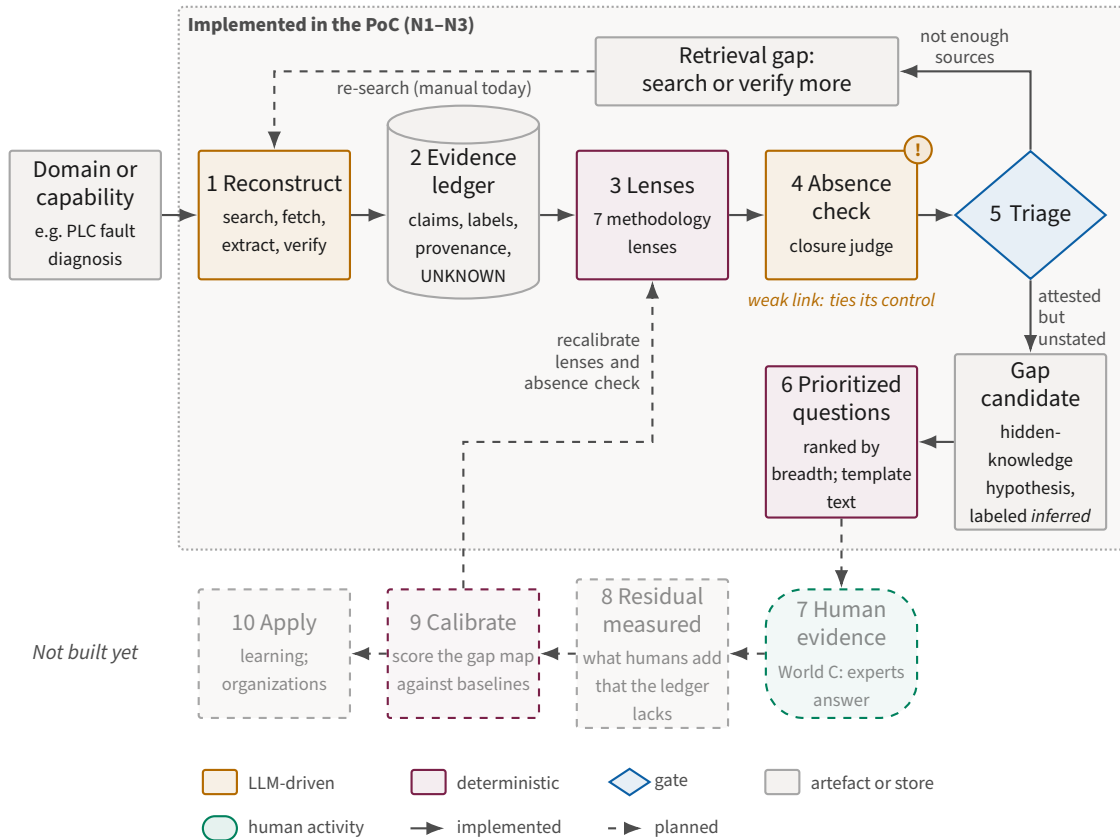


Figure 2: What is the whole approach, and which parts exist today? Steps 1–6 (solid) are implemented in the PoC; steps 7–9 (dashed) are future work. Triage splits the output into retrieval gaps, listed for further search, and gap candidates, which become hypotheses labeled *inferred* and expert questions. The absence check (step 4) is the weak link.

1–2 Reconstruct into an evidence ledger

Fetch public sources, extract claims, locate each quote in a hashed snapshot, and verify it with a model of another family. Each claim carries one of six epistemic labels; unknown marks a slot searched and not covered.

3 Lenses

Seven lenses *fire* when sources attest a construct but not its content, for example rival causes with no sign that separates them.

4 Absence check

A judge reads the sentences a lexical retriever pulls from the ledger (up to 8 verified and 4 unverified per probe) and returns *closed*, *partial* or *open*; closed firings are dropped.

5 Triage

A **retrieval gap** (nothing found, one independent source, unverified text only, answered by a sibling run, or an unreadable verdict) calls for more search, not an expert. A **gap candidate** has at least two independent sources and an element the absence check did not find fully stated. It becomes a hidden-knowledge hypothesis labeled *inferred*.

6 Questions

Template-filled (no model writes them), with a predicted knowledge type and a human channel; ranked by a breadth rubric, not by expected information gain.

7–9 Future

Expert answers measure the residual, score and recalibrate the gap map, and only then feed

LENS	SOURCE CONSTRUCT	LOOKS FOR	PREDICTED TYPE	ASK VIA
DISC	CTA (CDM): cue as discrimination	boundary of a judgment term	cue perceptual or collective	contrasting cases; rated vignettes
DIAG	PARI, ECD: rival causes	a sign that tells the causes apart	interpretation, cue relational	CDM on a recalled case, then contrasting cases
SEL	GOMS, HTA, KC: selection rule	situation-to-option mapping	decision relational	CDM: options and basis of the choice
HEDGE	CBR, KLI: when not to apply	the exception condition	decision relational or collective	CDM hypotheticals; boundary vignettes
GUARD	Automaticity: self-check habits	how the error is noticed	check, metacognition automated	observation first; probe is weak
WHY	Szulanski: causal ambiguity	the reason (rationale)	rationale relational	targeted confirmation question
RESULT	PARI: result interpretation	what a result means and where it leads	interpretation, expectancy relational	CDM on a recalled case; process tracing

The literature supports the constructs in column 2, not the lenses themselves.

Figure 3: Which research method gives which methodology lens, what missing element each lens looks for, its predicted knowledge type, and the human channel that would ask about it. None of the lenses has been validated against a human judgment of where expert knowledge is missing.

instruction or organizational uses.

H HYP. If a lens fires, a working absence check does not find the missing element, and the attested side has broad support, then practitioners are more likely there than elsewhere to hold knowledge the text lacks. The PoC builds the mechanism; it does not test this hypothesis.

4.2 What each tradition contributes

Figure 3 shows which research tradition gives which lens, the missing element each lens looks for, and the human channel that would ask about it. **L LIT.** The traditions differ in strength. That experts omit decisions, cues and interpretations is established in direction, from small studies in single domains (Section 2). CTA-based instruction beats other ways of choosing training content in a meta-analysis of few studies ($g = 0.871$; Tofel-Grehl and Feldon 2013). The Critical Decision Method, PARI, Evidence-Centered Design and knowledge components are established methods (Hall et al. 1995; Klein et al. 1989; Koedinger et al. 2012; Mislevy et al. 2003). Causal ambiguity rests on one large study of 271 observations (Szulanski 1996); the tacit knowledge typology is conceptual, not empirical (Collins 2010; Polanyi 1966); and reporting bias is shown for commonsense facts, not measured in expert text (Gordon and Van Durme 2013). **I INT.** The literature therefore supports the *constructs* the lenses use. It does not support the lenses: no cited study tests whether a detector built on these constructs finds real expert knowledge in text (Extended Technical Report, Section 3). Every mapping from a construct to a lens is our interpretation, and every lens output carries the label *inferred*.

4.3 Closest prior work, and what may be new

L **LIT.** Our search, to September 2026, found precedent for every step. Torre et al. (2020) check GDPR privacy policies for content types that a conceptual model of the regulation requires; on 24 unseen policies they found 45 of 47 incompleteness issues. That is typed absence detection in one of our own domains, but the reference fixes in advance what should be present. Anthonio et al. (2022) find underspecified phrases in instructional text, with later human edits as gold. DRMiner mines design rationale from issue trackers (Zhao et al. 2024). Luitel et al. (2024) flag terms missing from requirements, and OntoAgent picks interview questions from an ontology (Jin et al. 2026).

I **INT.** Each works on one document or interview against a reference that already contains the missing content. None predicts what practitioners know beyond all written sources, or routes a gap to an elicitation channel. Table 2 places the main elements.

Table 2: Condensed novelty ledger; the full ledger is in the Extended Technical Report, Section 4.3. *Adaptation:* a known idea on a new unit or purpose.

Element	Category	Precedent or reason
Claim ledger: verbatim spans from hashed snapshots, verifier of another family	Known combination	Min et al. (2023) and Rashkin et al. (2023)
unknown only for a searched, uncovered slot	Adaptation	Abstention and completeness (Razniewski et al. 2024)
Methodology lenses as absence detectors	Adaptation	Typed absence detection exists (Anthonio et al. 2022; Torre et al. 2020); new here only the source of the types and the target. Validity not shown
Knowledge type and elicitation channel per candidate	Adaptation / known method	Taxonomies and channel choice exist (Collins 2010; Klein et al. 1989)
Retrieval gaps routed to search, not to experts	Adaptation	Retrieve-or-ask split in Taranukhin et al. (2026)
Synthetic output as predictor only, never criterion	Known method	Limits of simulated samples (Bisbee et al. 2024)
Scoring a gap map against published CTA results	Potentially novel (uncertain)	No such comparison found; gold sets are small
Whole chain, domain-neutral, evidence status carried end to end	Potentially novel (uncertain)	No single system found; commercial tools may not publish methods

I **INT.** The most defensible claim is the combination: a provenance ledger with an evidence status per claim, expertise constructs as typed absence detectors, a knowledge type and channel per suspected gap, and search failures kept apart from expert questions. It holds only as a proof of concept. The evaluation design that scores a gap map against published CTA results is a stronger novelty candidate than the system, but it has not run.

5 What was built

The PoC had one narrow engineering goal: a working, checkable chain from a named task to ranked questions for an expert, using only public evidence and zero participants. It was built inside the NOW program, which allows no participants, no proprietary data and no live interviews; its one human input, a blind second coder, has not yet been used. **I** **INT.** The constraint came from the project's history. An earlier physics study, now frozen as Validation Track A, built a complete, tested instrument, but every one of its gates needed recruited physicists or students before any result could be read (Extended Technical Report, Section 5).

5.1 Architecture

o OBS. The PoC is three Python packages in a strict one-way dependency chain, checked by an architecture test: `residual/` (268 tests), `reconstruct/` (416) and `gapmap/` (172), all offline with fakes, never a live model (Extended Technical Report, Section 6). `residual/` defines the typed evidence model. `reconstruct/` populates a ledger for one (domain, task) pair from the open web, in one strictly ordered call with no agent loop: plan 3–6 areas with queries; search; fetch, normalize and hash each document once into a content-addressed store; extract candidate claims; *locate* each quote verbatim in the normalized text (or produce no evidence at all); cluster documents into independent sources; and *verify* support with a model of another family. `gapmap/` reads one ledger and proposes gap candidates; it never writes a hypothesis back as a claim.

o OBS. Planner and extractor are `anthropic/claude-haiku-4.5`; the verifier is `openai/gpt-5.4-mini`. The code refuses a verifier from the extractor’s family, so no claim gains a criterion label by self-verification. The closure judge is `qwen2.5:7b-instruct`, run locally at zero cost and chosen because its family differs from both. Its verdicts are cached by prompt hash, so the gap map replays offline and byte-identically; only an explicit flag calls the model. A budget check stops before overspending and writes the outputs as incomplete rather than truncating silently.

5.2 The evidence model

o OBS. The atomic unit is the `ClaimRecord`. Each claim carries exactly one of six epistemic labels (Table 3). The label is a computed property, never a stored field: nothing in the code can set it by hand. A derived claim takes the weakest label among its premises, so a chain of reasoning cannot bootstrap a strong label from a weak one. The evidential gate refuses any `synthetic_extrapolation` input, and anything else outside the three criterion labels, before the gated function runs. This is the code form of the project’s central rule: **synthetic output may be a predictor, never a criterion.**

Table 3: The six epistemic labels (exact enum values). Only the first three can act as a criterion.

Label	Meaning	Criterion
<code>observed_human_evidence</code>	Supported by a human-record (World C) source	Yes
<code>literature_supported</code>	Supported by a verified span in a public (World A) source	Yes
<code>organisational_artefact_supported</code>	Supported by a verified span in an organization’s own (World B) artifact	Yes
<code>inferred</code>	Derived from other claims, with no direct support	No
<code>synthetic_extrapolation</code>	Produced or supported only by a model	No
<code>unknown</code>	Searched, fetched and fully examined by another model family; nothing found	No

Provenance. **o OBS.** Every fetched document is normalized once and hashed. A selector names an exact character span inside that hash, and its text is the literal slice of the snapshot, never the model’s own quotation. The chain is *snapshot hash* → *re-sliced span* → *claim*, so every evidence item can be re-checked offline with no model call. A claim whose quote cannot be located gets no evidence and stays in the ledger under `synthetic_extrapolation`; it is not

dropped. Memorized model content can therefore enter the ledger, but only under a label no lens or gate reads.

UNKNOWN, independence and contradictions. o **OBS.** A claim with no evidence enters under unknown only when its slot was searched, fetched and fully verified with nothing found. Weaker states (one source, unverified, never fetched) are tracked separately, because a slot never searched is a retrieval failure, not a silence of the domain. Corroboration counts independent source clusters, not documents. When two claims from different clusters conflict, the ledger stores a contradiction record rather than merging them.

Evidence worlds. o **OBS.** Public (World A), organizational (World B) and human (World C) evidence are a typed field on each source, with validators. **Only World A is populated in the PoC:** World B has no ingestion adapter and World C has no data. The schema, the gap-map feature set and the label code are hash-frozen together; a change needs a deliberate re-freeze and a commit saying why.

5.3 N0–N3 and the verdicts they read

o **OBS.** Every NOW item is pre-registered to end in a continue, change or stop reading of an experiment, never in “the build finished”.

- **N0** (reset; DOI hook, Track A frozen): done, gate passed.
- **N1** (evidence model): done, gate *continue with 2 additions*, at the registered limit of two.
- **N2** (reconstruction): engineering complete; registered research verdict **INCONCLUSIVE** (Section 6.1).
- **N3** (gap map): pipeline complete; hidden-knowledge prediction **not demonstrated**; its program gate (Δ AUROC against gold) has not run.

i **INT.** Under the plan’s rules, N2’s inconclusive verdict would have blocked N3. The owner made one recorded exception and treated N2’s engineering milestone as enough to start N3. That was a softened gate, softened by the person whose program depends on it, and no one outside the project checked it. The validation program therefore adds an external gate reader (Section 10).

6 Results

6.1 N2: reconstruction on the live web

N2 ran three experiments. E-LIVE checks that the pipeline runs on the live web without fabricating or mislabeling; it has no gold and no threshold. E-PLANT and E-ABST are pre-registered, gated experiments that compare the gated pipeline with an ungated raw model.

E-LIVE. o **OBS.** Two domains ran with the same models: PLC intermittent-fault diagnosis (technical, procedural) and GDPR data protection impact assessments (legal, normative). A post-fix PLC rerun crashed on a PDF encoding defect (0 claims). Table 4 gives the completed runs.²

² `research/n2/e_live_report.md`, §3 and §5; recomputed from each run’s `sidecar.json` and `ledger.json`.

Table 4: E-LIVE per-run counts. “Located” counts claims with at least one verbatim span; “Evidence” counts the spans, which is larger where a claim holds several. “Verified” counts claims the verifier accepted. “Unknown” claims are slots searched, fetched and fully verified with nothing found. GDPR v2 is defect-compromised: 140 of its 366 evidence items kept a pending verdict after silent verifier failures, so it is not admissible for N3.

Run	Sources	Extracted	Located	Evidence	Verified	Unknown	Cost
PLC v1	35	370	337	337	321	5	\$0.679
GDPR v1	20	259	204	206	195	14	\$0.500
GDPR v2 (not admissible)	25	523	351	366	207	15	\$1.002

o OBS. *Exact spans.* All 909 evidence items of the three ledgers re-slice exactly from their snapshots (337 + 206 + 366); a separate sample of 71 re-sliced 71 of 71, checked by an AI inspector, not a human. The claims are a different matter: of 1,169 claims, 418 are labeled `synthetic_extrapolation` (49, 64 and 305: 260 with no locatable span, 158 with a span the verifier did not accept) and 34 unknown, so 452 (39%) have no supporting span. *Verifier error.* On sampled claims the verifier’s main error is scope drift, not fabrication: 30% on PLC v1 (Wilson 95% CI 15–52%, $n = 20$) and 12% on GDPR v2 (4–30%, $n = 25$, in-sample). GDPR v1’s 14 unknown slots concentrate in `failure_mode` (5 of 5 areas): regulator text states obligations, not how an assessment goes wrong in practice; **i INT.** this fits the kind of gap the gap map targets, but it is not validated.

o OBS. *Decoys.* Supported claims were mutated (negation, a changed number or scope) and checked again against the original span. The verifier accepted 7 of 20 decoys on PLC v1 (35%, 18–57%) and 3 of 20 on GDPR v1 (15%, 5–36%). Three PLC scope decoys only dropped a conjunct, so the span still entailed them; without them the rate is 4 of 17 (24%, 10–47%). **i INT.** The label `literature_supported`, which every lens reads, means “a locatable span in a public source that a verifier model accepted”, not a fact established by the scientific literature. That verifier accepted between a quarter and a third of mutated PLC claims, and its errors pass into lens firing and candidate generation: a false “supported” claim can seed a gap candidate.

o OBS. *Defects found live.* Running on the real web surfaced eleven defects that 416 passing offline tests had missed, among them verifier failures hidden behind a complete: `true` flag (Extended Technical Report, Appendix D). None of the fixes was re-checked live, and the current cross-verify and decoy stages have never run live.

E-PLANT and E-ABST. E-PLANT gives the raw model a false planted value among about 12 real pages that state the truth, and asks whether it repeats the plant. Its validity floor assumed the ungated model would adopt plants on more than 60% of targets, a figure taken from ClashEval (Wu et al. 2024). That figure holds only for two models, with the planted document as the only context, a condition E-PLANT did not reproduce. E-ABST asks the raw model real but private questions and counts false answers where it should abstain.

o OBS. OBSERVED IN THE PoC

N2’s registered verdict: INCONCLUSIVE. The ungated model adopted $a_U = 5$ of 24 plants (21%, Wilson 9–40%), below the registered validity floor of 8. By the pre-registered rule this closed the Continue route before any gated arm was scored; it is a result about the test’s sensitivity, not about gating. In E-ABST the raw model answered only 5 of 24 private items and abstained on 19 (79%), so the required 20-point margin was reachable only in a narrow corner. The gated arms were never scored. **No statement about whether gating reduces planted falsehoods or false answers is supported.**

o OBS. *Budget stop.* N2’s total spend was about \$4.82 against a \$5 weekly key limit (E-LIVE \$2.374, E-PLANT \$1.953, E-ABST \$0.495), while the three experiments’ registered caps summed to about \$21. The post-fix live re-check (E-LIVE v3) stopped at its first call on the provider’s key limit. The owner chose not to raise it. **i INT.** Budget headroom is part of the design: limits must be checked against the sum of caps before a run sequence starts, and a “beats the baseline” gate must be calibrated on a pilot before it is registered.

6.2 N3: the gap map

Lenses. **o OBS.** Seven lenses fire over criterion-labeled claims only. They read the extractor’s paraphrase of each claim, not the verbatim span, and three of them depend on knowledge types the extractor assigned. Lens firing tracks domain texture as designed: DIAG fires only on PLC, the diagnostic domain; DISC and HEDGE fire mainly on GDPR, the normative one (Table 5). This shows that lenses distinguish texture, not that they find hidden knowledge.

Table 5: Lens firing per ledger. “Promotional excluded” counts candidates dropped before judgment by a domain denylist tuned in-sample.

Ledger	Fired	By lens	Promotional excluded
PLC	56	WHY 24, RESULT 19, DIAG 9, SEL 2, DISC 1, GUARD 1	42
GDPR v1	51	RESULT 21, DISC 13, WHY 8, HEDGE 5, SEL 4	8
GDPR v2	48	DISC 14, SEL 11, RESULT 9, WHY 9, HEDGE 5	1

The absence check. A fired lens only shows that a construct is named. Deciding that its content is really *unsaid* is the absence check. The PoC tried two versions.

o OBS. *Lexical closure* scored a candidate closed if another claim shared enough word stems. Against a null of 200 random-donor draws, the observed closure counts fall within or below the null’s 5–95% interval on every ledger: PLC 22 against a null mean of 33.48 (26–41), GDPR v1 11 against 12.735 (8–17), GDPR v2 11 against 8.08 (5–12). Lexical closure did no better than chance.

o OBS. *Semantic closure*, the final version, asks the local judge to decide from retrieved sentences whether the missing element is closed, partial or open; retrieval also injects the seed span’s own sentences. The *mismatched-evidence control* (earlier drafts called it the “fair control”; code field `mismatched_evidence_control`) keeps the seed’s sentences in both arms and swaps only the other-source sentences for those of a donor: the topically nearest candidate of the same lens. Counting partial as closed, the judge’s rate is the same in both arms: **PLC 0.889, GDPR v1 0.961, GDPR v2 0.958** ($n = 54, 51, 48$). Every tested lens in every ledger is flagged closure-uninformative (Figure 4).³

³ Each `research/n3/*/gapmap.json`, field `checks.mismatched_evidence_control`.

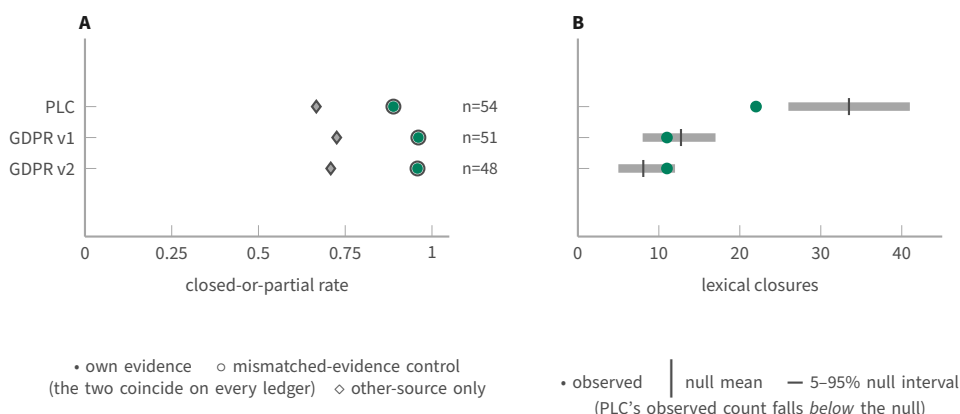


Figure 4: Does the absence check beat its mismatched-evidence control? It does not. Panel A: closed-or-partial rate on a candidate's own evidence and on the control (other-source sentences from a donor), by ledger; the rates are equal. With the seed and its source removed (other-source only), the rate falls but stays high. Panel B: observed lexical closures against the donor null (mean and 5-95% interval of 200 draws).

Four qualifications keep this in proportion. In 32 of the 153 pairs the two arms sent the judge an identical prompt, so equality there holds by construction. Counted on `closed` alone, own versus control is 0.30/0.33, 0.65/0.61 and 0.58/0.52, still far below the 0.20 difference the check requires. The own rate is 1.0 on 11 of the 14 tested lens-`ledger` cells, so the test has little room. And an arm with the seed and its source removed still closes or partly closes 0.667, 0.725 and 0.708 of candidates, against 0.889, 0.961 and 0.958 with them. **I INT.** The reading is narrow: under this control, the judge did not tell a candidate's own additional cross-source evidence from the donor's. The test does not say why. Judge leniency, the weight of the seed evidence both arms share, donor evidence that is genuinely relevant (the donor is the nearest same-lens topic), or a control too insensitive to show a difference could each produce the tie. **O OBS.** One warning sign points at the seed: in the pre-final maps, 17 of PLC's 22 closures cited only the seed or its own source.⁴ **I INT.** So the problem may be not only retrieval of the wrong support: the formulation may let the sentence that triggered the lens give a lenient judge enough to close its own gap. Every candidate reads "a lens fired here", not "experts know something here". **O OBS.** An adversarial AI review of the pre-final maps found this weakness; the first built-in control had been a straw man that "passed". The mismatched-evidence control now runs on every replay.

An exploratory comparator: Jev. **O OBS.** After the maps were final, a small spike (\$0.45 in total) tested whether a low-cost decision model, Jev (TypeSafe `jev-1.13`, which returns probabilities for typed questions and no text), gives a useful alternative reading of the committed N3 output. It consumed committed artifacts only: the text of 61 stored candidate and retrieval-gap records from the three `gapmap.json` files, and all 641 closure-judge prompts, recovered byte for byte by an offline replay of the judge cache. Nothing in the gap map changed.⁵ **Signal.** Asked per record whether the missing element is practice-only know-how, Jev gave consistent scores over two paraphrased questions ($\rho = 0.91$), related to source density ($\rho = 0.28$ pooled, 0.51 on PLC) and to the N3 score (0.18; 0.42 on PLC). But two word-length statistics also passed the pre-set rule. Replacing every word with a random word of the same length still left pooled $\rho = 0.61$ with the original score, so the pre-set form rule reads **NO SIGNAL**. (The pooled figure is inflated by an

⁴ `docs/lessons.md`; `gapmap/src/gapmap/checks.py`.

⁵ `research/spikes/jev/RESULTS.md` and `analysis*.json` there.

unchanged field line; a post-hoc within-domain reading, 0.35–0.47, would be INCONCLUSIVE.) The PLC density link fell from 0.51 to 0.06 on the meaningless text. *Judge*. Jev’s agreement with the stored qwen answers failed its pre-set rule ($\kappa = 0.19$). A second reference, Sonnet 5.5, was then chosen. On 100 prompts from 68 seeds, Jev agreed with it at $\kappa = 0.53$ against qwen’s 0.27 (difference 0.25, 95% CI 0.10–0.40 resampling prompts; 0.32 when qwen got Jev’s instructions), at about hosted qwen’s cost per call. **I** INT. That is agreement with one model, not accuracy, and Jev was not run on the mismatched-evidence control, so whether a different judge would break the tie is untested. Jev is an exploratory comparator. It does not detect the Human Knowledge Residual. In the program it is one more judge in V1 and one more pre-registered baseline in V2 and V5, never a gate or a criterion.

Triage, ranking and the display cap. **O** OBS. An unclosed firing becomes a retrieval gap (RG-UNK: nothing found; RG-SINGLE: one independent source; RG-UNVER: closed only by unverified text; RG-SIBLING: closed in a sibling run; RG-UNDECIDED: unreadable judge answer), whose action is more search, or a gap candidate (HYP): an open or partial record with at least two independent sources. A HYP record has three tiers: observed evidence, the inferred gap, and a hypothesis whose label is fixed to inferred. Candidates are ranked by a breadth rubric (independent sources, step breadth, penalties for partial and promotional seeds), not by expected information gain. After merging near-duplicates, a display cap keeps at most 12 records, 3 per lens and 4 per area. On PLC it hides 15 of 24 merged candidates (10 RESULT, 5 WHY). Questions are filled from fixed templates with up to two verbatim quotes; no model writes them.

Table 6: Final map counts. GDPR v2 is shown only for the cross-run check. RG-UNK is a different quantity from N2’s unknown claims (Table 4).

	PLC	GDPR v1	GDPR v2 (not admissible)
HYP after merge (uncapped)	24: RESULT 13, WHY 8, DIAG 2, GUARD 1	1: RESULT	6: RESULT 3, DISC 1, HEDGE 1, WHY 1
HYP shown after the cap	9: RESULT 3, WHY 3, DIAG 2, GUARD 1	1	6
Closure state of shown HYP	3 open, 6 partial	1 partial	6 partial
RG-SINGLE / UNVER / SIBLING / UNK	11 / 0 / n/a / 10	12 / 4 / 2 / 28	3 / 6 / 7 / 44

The negative results, stated plainly.

- **O** OBS. **13 of the 16 displayed candidates are partial.** The two PLC DIAG records read open only because the DIAG rule counts a partially signed rival cause as unsigned; only one displayed candidate (PLC rank 7, a WHY record) has no partial hit. **I** INT. The control counts partial as closed, so most surviving hypotheses sit in a state the map’s own control would call closed.
- **O** OBS. **Precision by adversarial judgment: 3 of 19 strict (8 of 19 lenient), on the pre-final maps only** (Wilson 5.5–37.6% and 23.1–63.7%). One AI reviewer judged in-sample, with no human coder and no gold; the per-record tally is not committed, and the final maps were not re-tallied. It is a sanity signal, never a measured precision.
- **O** OBS. **PLC’s ranking follows source density** (Spearman $\rho = 0.91$; overlap of the top ten with the ten highest-density candidates $J_{10} = 0.58$). The anti-renaming check tests only the low-density direction ($J_{10} = 0.00$). On GDPR, ρ is 0.09 and 0.16.

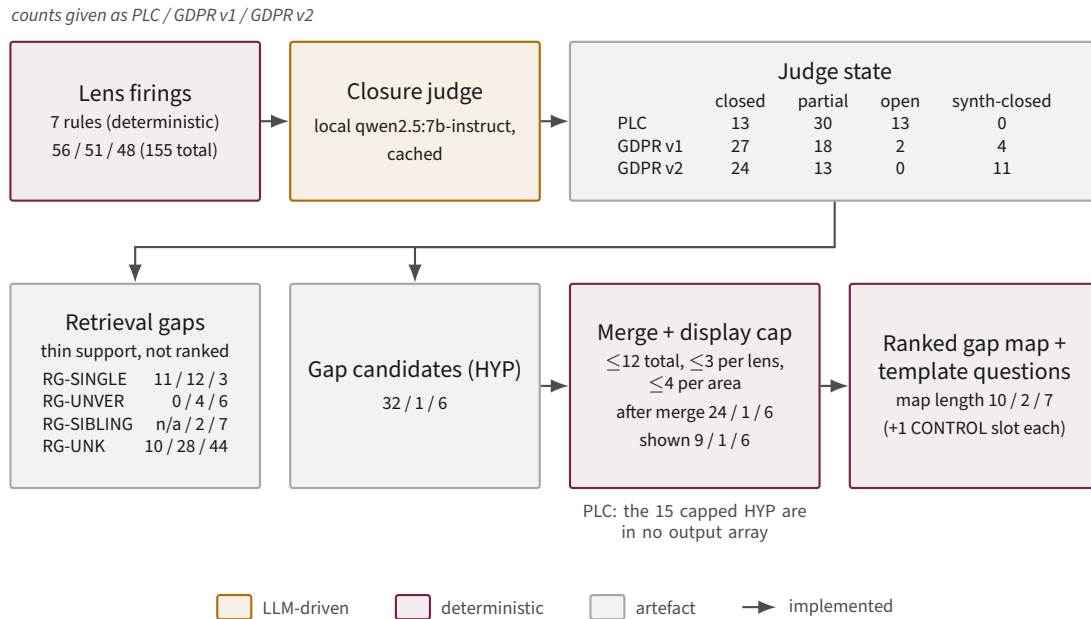


Figure 5: How lens firings become retrieval gaps, gap candidates or nothing, with counts per ledger (PLC / GDPR v1 / GDPR v2). After triage, near-duplicate HYP records are merged (PLC 24) and a display cap keeps at most 12 (PLC 9 shown). Most shown candidates are *partial*, not *open*.

- o **OBS.** **Cross-run stability is low.** Of GDPR v1’s 13 unclosed lens records, 3 have a counterpart in the independent v2 run; of v2’s 9, 3 have one in v1. Every counterpart is *partial*. i **INT.** The runs are dominated by different sources, so the instability traces to retrieval.
- o **OBS.** **A judge-parsing defect was found by review, not by tests.** Unparsed judge answers had silently defaulted to *open* (reported as 24 of 103, 62 of 150 and 60 of 147 responses). The tests used a fake judge that always answered in format. Unparsable answers now become RG-UNDECIDED.

6.3 One complete trace, and its weaknesses

Figure 6 follows PLC’s top-ranked record from source to question. The seed span, from a public troubleshooting page, reads “Loose terminals cause many intermittent faults. A flashlight reveals problems in dark panels that you miss otherwise.” The claim is `literature_supported`. Seven independence keys share the topic’s stems (two are vendor product pages), and the DIAG lens found five rival causes (ground loops, aliasing, installation versus software, VFD/contacter noise, pull-up/clock timing). Of 34 verified sentences retrieved, the judge found none that fully states a sign separating any rival, and it marked 5 claims as *partial*. The DIAG rule counts a partially signed rival as unsigned, so the record reads *open*. The template question asks: “Which cause did you suspect first, and what did you see, hear or measure that let you rule the others out?”, to be asked as CDM probes on a recalled case, then contrasting cases.⁶

o **OBS.** *Weaknesses.* The trace was chosen by the author after the maps were final (the top-ranked record of each map). The question’s anchor is the extractor’s assertion cut at 80 characters, so the question is not a grammatical sentence. The hypothesis template says the judge “found a sign for the following” and then marks every item “(no sign found)”. The *open* state overstates

⁶ `research/n3/plc/gapmap.json, map[0], gap g-59164a9db5bd.`

the evidence, since the judge found partial signs. PLC’s second record repeats the fragment defect, with rival causes such as “array subscript out of range.” Such a candidate should not reach an interview unedited. **I** INT. The trace shows that the mechanism runs from a real span to a fixed-frame question, and where model-written text enters. It cannot show, without a human answer, that the question would surface knowledge a practitioner holds.

7 What the PoC shows and does not show

Table 7 lists only what the repository artifacts support, and what they do not. The claims are about the mechanism: that it runs, keeps its evidence traceable, and carries checks that expose its own weakness. None is a claim about hidden expert knowledge. Evidence rows and artifact paths are in the Extended Technical Report, Section 12 and Appendix A.

Table 7: Part A: what the PoC demonstrates. *Strong*: direct evidence checked by code or a test, or seen in more than one run. *Moderate*: one run or in-sample. Part B: what it does not demonstrate, and what evidence would be needed.

A. Demonstrated O OBS.		
Claim	Main limitation	Strength
The evidence model enforces its rules in code: computed labels, gates that refuse synthetic or unknown criteria, a hash-frozen schema	Tested offline; no World B or C evidence has passed through	Strong
Reconstruction runs end to end on live sources; every evidence item re-slices from its snapshot	One complete run per domain; 452 of 1,169 claims have no supporting span and are labeled so	Strong
Cross-family verification and UNKNOWN slots operate	Operation, not accuracy: verifier error 12–30% by one AI inspector; cross-verify never ran live	Moderate
Live runs expose defects offline tests miss, at small cost	Eleven fixes never re-checked live	Moderate
The gap map runs end to end, keeps retrieval gaps apart from candidates, replays byte-identically	A display cap hides most PLC candidates	Strong
The built-in checks expose the pipeline’s main weakness	They show the absence check does not work yet, not that it can	Strong
B. Not demonstrated		
Claim	Why not, and what would be needed	
That gap candidates are real tacit knowledge (H2)	No human has seen a candidate; no gold. Needs human-coded gold in held-out domains and a pre-registered test	
That the absence check is informative	Ties its mismatched-evidence control everywhere. Needs agreeing human closure labels and a judge that beats the control (V1)	
A precision for the gap map	Only one AI reading of pre-final maps. Needs two blind human coders on frozen maps	
How many candidates the pipeline produces	Maps are capped; no area score. Needs a pre-registered area score over the uncapped set	
Robustness to planted falsehoods; abstention	Gated arms never scored. Needs a calibration pilot, budget headroom, scored runs	
Reconstruction quality against gold	“Supports” is a model’s judgment; lenses read paraphrase. Needs expert-coded claims	
Generalization and stability	Two domains, English web, in-sample tuning; two GDPR runs disagree. Needs held-out domains, a dated frozen corpus, repeated runs	
That the ranking is not source density	$\rho = 0.91$ on PLC. Needs density and area size as headline baselines	
Expected-information-gain question selection	Not implemented; ranking is a breadth rubric	
Value for learners or organizations; World B or C	Zero participants by design; typed fields only	

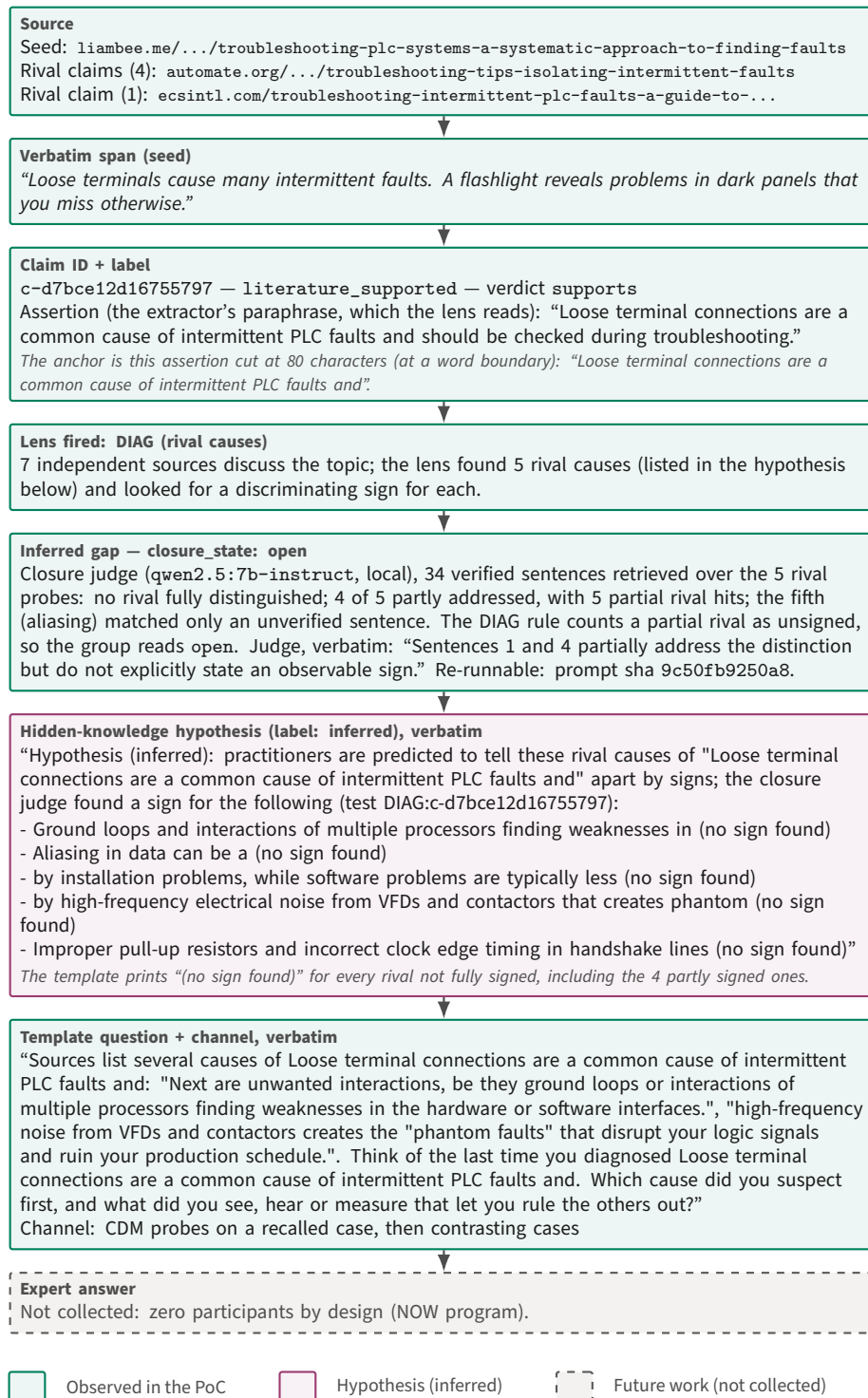


Figure 6: One complete real trace, end to end: seed span and claim, DIAG lens with five rival causes, the closure judge's verdict (no rival fully signed; 5 claims partly signed, counted as unsigned, so the record reads open), the hypothesis and the template question. The lens fired; the absence check found the element at most partly stated. No expert has answered the question.

■ FINDING

The PoC shows a working, checkable mechanism. It does not show that the mechanism finds real hidden knowledge. This report does not claim that the system identifies tacit knowledge, that the absence check found an element stated nowhere, that the ranking reflects real gaps rather than how much is written, that the pipeline resists planted falsehoods, that N2 passed or failed, or that every claim in the ledger is tied to a source span.

8 Interpretation and threats to validity

I INT. INTERPRETATION

The bottleneck is absence detection, not reconstruction and not lens firing. Reconstruction with provenance runs, and lenses fire in the expected places. What has no valid answer yet is “is this element really unsaid?” Every quantity after step 4 of the central model depends on it: which records become candidates, how they rank, which questions reach an expert, and any future score against gold. It must be validated against human closure labels before any gold test, or that test would score lens firing plus density, not the gap map.

I INT. Two features of the pipeline make the failure likely. Lenses fire on word stems in the extractor’s paraphrase, so they cannot see a paraphrase of the missing element and often fire where the seed already states part of it. And most displayed candidates are *partial*: they mark places where the text says *something* about the element. Three caveats keep this from being a verdict on absence detection as such: the judge is one small local model, a stronger judge is untested, and the ceiling leaves little room for any difference.

I INT. **Thinness is not the residual.** An area can be thin because the knowledge is unwritten, or because retrieval was poor. O OBS. The PoC ledgers show much of the second kind: in GDPR v1 one independence key carries 194 of the ledger’s key references and there are only three distinct keys; in PLC v1, 11 of 35 fetched “sources” were error pages. I INT. On PLC the ranking is close to “how much is written about this topic”, which could look predictive for the wrong reason, since experts may also have more to say about central topics. So density and area size must be headline baselines, and the test must be conditional on them. The map is also partly a property of retrieval: two runs of one task gave mostly different candidates. Any gold test needs a dated, frozen corpus and repeated runs. Finally, lexicons, denylist and stop-anchors were tuned on the same ledgers, so any number measured on them is optimistic; tuning did not rescue the absence check.

■ FINDING

The PoC neither demonstrates nor refutes H2. It does not demonstrate it: no gold, an absence check that ties its control, and a low, in-sample precision signal. It does not refute it: the registered stop rule needs held-out gold, a dated corpus, a frozen configuration, δ , an area score and a second coder, and none exists. A negative result from a component not shown to measure its target says nothing about the target. The correct reading is “prediction not demonstrated, and the current absence check is not yet a valid instrument”, not “the human residual is unpredictable”.

What would change our mind. *Toward H2:* in blind prospective elicitation the lower 90% bound of ΔAUROC is above 0 and the point estimate at least δ . *Preconditions, not evidence for H2:* coders agree on stated / partly stated / not stated ($\kappa \geq 0.70$) and a judge beats the mismatched-evidence

Table 8: Main threats to the PoC’s conclusions, what mitigates them now, and what the validation program does.

Threat	Effect on our claims	Now	Program
What a “gap” means	A firing plus an unclosed state may mark a paraphrase or a retrieval hole	Candidates labeled <i>inferred</i> ; retrieval gaps apart	Human closure labels (V1); gold tests
Judge leniency, seed dominance, ceiling	Own rate 1.0 on most lenses; seed kept in both arms; the tie has several possible causes	Tie reported with closed-only, other-source-only and paired figures	Human labels in both arms (V1)
Model paraphrase	Lenses and judge read the extractor’s assertion, not the span	Spans kept beside every record	Lenses on spans; types checked in V1
In-sample tuning	Positive numbers would be optimistic	Configuration hashed; tuning disclosed	Hash-freeze before gold (MS2)
Defaults that equal the finding	Unread answers became “open”; a first control could only pass	Parse fix; defects logged	Planted positives and nulls for every check
Two domains, English web	Nothing generalizes beyond PLC and GDPR	Claims limited to these runs	Held-out domains after the freeze
Non-reproducible retrieval	Web search, drifting pages and retiring models	Ledgers, snapshots, judge caches kept	Snapshot archives; dated corpora
Tiny n , clustered units	Precision, verifier error, cross-run counts rest on 19–71 items that share sources	Wilson intervals (which assume independence)	Cluster-aware intervals; sizes set after pilots
Single operator, owner incentive	One person designed, ran and read everything, and is the applicant	Committed artifacts; adversarial AI reviews	Non-owner blind coders; external gate reader
AI-assisted development	Agent agreement can look like corroboration	Every number traced to an artifact	Human coders and pre-registration decide

control and agrees with them; on pooled, dated, held-out CTA golds the frozen map beats density, area size and a plain-LLM ranking, which shows compatibility with CTA-derived representations. *Against H2:* coders cannot agree whether an element is stated ($\kappa < 0.70$ after one codebook revision); closure passes its human check yet the map adds nothing beyond density and area size; or the prospective upper 90% bound falls below δ . *Neither:* more runs on the same ledgers, agreement among AI reviewers, or a stop reading from a configuration whose absence check never passed its control.

8.1 Threats to validity

Table 8 lists the threats that could make the PoC’s readings too strong or too weak (full list: Extended Technical Report, Section 15).

I INT. The PoC code, its adversarial reviews and this report were produced with AI agents under the project lead’s direction. The reviews found real defects, including the parsing defect and the failed closure control, so they are useful as a search for errors. They cannot certify that errors are absent, and they cannot serve as a criterion.

9 The Human Knowledge Residual

The program has one target quantity that no one has measured: the Human Knowledge Residual (HKR). **I INT.** The term is not established. Its nearest analogues are unseen-item estimation (Chao 1987), recall of knowledge bases (Razniewski et al. 2024) and knowledge loss after developer turnover (Rigby et al. 2016).

H HYP. HYPOTHESIS

Fix a task scope D and an evidence base E : a set of sources found by a fixed search protocol and frozen at a date t . $\text{HKR}(D, E, t)$ is the set of knowledge items that (1) competent practitioners of D *demonstrably use*, corroborated by a second practitioner, a trace or an artifact outside E ; and (2) that E *does not contain*, as judged under a fixed codebook by blind coders or by a matcher that agrees with their adjudicated label (the adoption rule, $\kappa \geq 0.70$). We hypothesize that this set is not empty where practice matters, that it concentrates in cues, decision rules, checks and rationale, and that its location can be predicted from E alone better than by cheap baselines.

The definition is relative: a new source can shrink the HKR. Let R be what the reconstruction extracted and H the validated human items. The observed residual share is $|H \setminus R| / |H \cup R_{\text{valid}}|$, reported per knowledge type and area. **I INT.** $H \setminus R$ is only an upper bound: it splits into the residual $H \setminus E$ and reconstruction misses, items E states that the pipeline did not extract. Telling them apart needs each item checked against the frozen corpus, the “is it really unsaid?” decision the PoC’s absence check did not make reliably. So the residual cannot be measured more reliably than absence can be judged. The HKR itself is latent: a study observes only the *elicited residual*, the items of $H \setminus E$ that a fixed protocol (probe, time, experts, interviewer) brings out and corroborates, a lower bound on the HKR. An area is *residual-present* if it holds at least one validated, important elicited item; that is the label the gap map is scored against. The label errs one way: an area without such an item reads “none found by this protocol”, never “no residual”. Misses unrelated to the scores pull every AUROC toward 0.5; misses that track a score (absence is easier to confirm in thin areas, for example) can bias ΔAUROC either way.

I INT. Three quantities must not be blurred. *Thinness* is a property of the ledger alone; the PoC computed it. The HKR is a relation of H to E ; it is not yet measured. The *gap-map prediction* is an area score labeled inferred; none is built yet. Only the prospective study (V5) measures the elicited residual as defined. The pooled CTA study (V2) scores the map against published CTA gold lists, but the lenses were built from CTA constructs: freezing the map before gold is seen prevents item-level, not construct-level, dependence, so V2 screens and does not decide. The departure study (V3) measures realized knowledge loss, a different construct. **O OBS.** The accounting types exist and refuse shortcuts: they raise an error when an item lacks a human validity judgment and reject a matcher from the generator’s family. No estimator is implemented and no World C data exist.

10 The validation program

F FUT. Nothing in this section has run; rules not yet registered are marked “proposed”. The program asks for no money before its cheapest decisive test has run, and each tranche is released only by a gate that reads CONTINUE. The studies test one link at a time, and a failed link stops or reroutes everything above it: trustworthy inputs (L0, V0), an informative absence check (L1, V1), retrospective prediction on published gold (L2, V2), a measurable residual (L3, V4), prospective

F FUT. FUTURE WORK**Gate rules, proposed for pre-registration.**

MS1, closure verdict (end of Phase 0). CONTINUE if the coders reach $\kappa \geq 0.70$ on the gating split (closed versus not closed), corpus-level retrieval recall meets its floor (proposed 0.80), and at least one judge beats the mismatched-evidence control: its paired own-minus-control difference in the not-closed rate has a 95% lower bound above 0, adjusted for the number of judges and resampled by seed claim, and lies within ± 0.10 of the coders'. Otherwise STOP the automated absence check. κ shows that the label is reproducible, not that it is right.

MS3, screening gate for Phase 2 money (end of Phase 1). CONTINUE only if the pooled retrospective ΔAUROC has a point estimate $\geq \delta$, and its lower 90% bound is $> -\delta/2$, and the map adds value over density and area size (conditional AUROC within density tertiles $\geq 0.5 + \delta$). Otherwise STOP H2 funding. MS3 is not a verdict on H2 and has no inconclusive route.

MS5, prospective test of H2 (Phase 2). STOP, H2 falsified, if the upper 90% bound of the prospective ΔAUROC is below δ . CONTINUE, H2 survives, if the lower 90% bound is above 0 and the point estimate is at least δ ; this rejects “no gain” but does not establish a gain of at least δ , which only a lower bound above δ would. Any other reading is reported as not supporting H2 and ends H2 funding.

prediction (L4, V5, the prospective test of H2) and practical value (L5, V6–V8; V6 tests H3). **I INT.** L1 comes first: with an uninformative closure step, a gold test measures “a lens fired” plus density. The departure study V3 is off this chain; it tests H-OSS.

Who reads the gates. An external gate reader, an independent statistician or an advisory board with one, reads MS1, MS3 and MS5 against the hash-committed pre-registration, with authority over them. The reader is not a co-applicant and is not paid from a tranche the reading releases. Any owner exception, at any gate, needs the reader’s written sign-off, or the reading is void. **O OBS.** No such reader existed during the PoC, and none is named yet.

10.1 Rules common to every study

F FUT. *Two-step freeze.* The judge is chosen at MS1. Then MS2 (about month 4 of Phase 1) hash-commits the pre-registrations, lexicons, thresholds, the adopted judge, the area score and the list of eligible golds (Nosek et al. 2018). Only then is gold acquired and are held-out domains reconstructed. *Area score.* Every decisive test ranks areas, so Phase 1 builds a fixed aggregation (for example the maximum or a noisy-OR) over the *uncapped* candidates in each area, chosen on pilot data only. *Density.* Candidacy and rank are built from source counts, so a plain gain over density is close to testing a variable against itself. The primary incremental analysis is the area score’s AUROC within density tertiles. *Domains and dated corpora.* PLC and GDPR serve only for debugging, V1 and an unanalyzed V5 pilot. Decisive studies use held-out domains on a corpus dated before the criterion (archive captures and publications dated before the cutoff), with a per-gold blocklist and a memorization probe on every model in the chain (Golchin and Surdeanu 2023; Sainz et al. 2023). *Judges.* No LLM judge is a criterion. One is adopted only from another family than the generator, with $\kappa \geq 0.70$ against the coders’ adjudicated label (the adoption rule), and re-checked on each new text type before gold is unsealed, which also offsets picking the best of a panel; where it fails, closure there is human-coded.

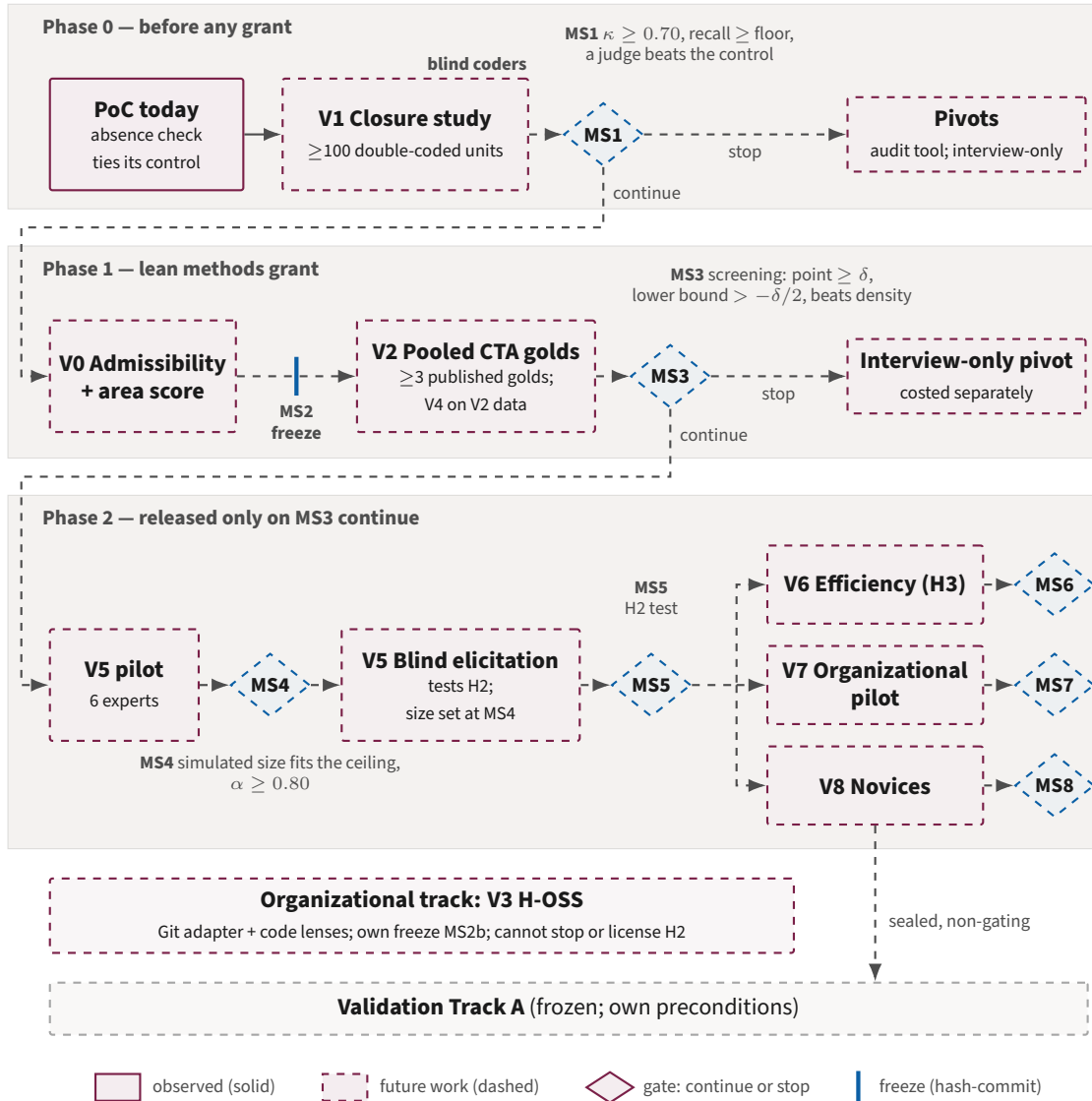


Figure 7: Which studies, in which order, turn gap candidates into validated predictions, and which gate releases which money? Phase 0 (V1) ends at MS1. Phase 1 (V0, the MS2 freeze, pooled V2, V4) ends at the MS3 screening gate. In Phase 2, V5 tests H2 at MS5 after its pilot gate MS4; only then do V6 (H3), V7 and V8 follow. V3 tests the separate hypothesis H-OSS. Stops lead to pivots. Track A runs on a separate rail. All elements except the PoC are future work.

F **FUT.** *Statistics.* Area-level AUROC is compared by DeLong’s method or a cluster bootstrap (DeLong et al. 1988; Obuchowski 1997), with the strongest baseline chosen inside each resample; MS5 pairs a one-sided test against δ , which can falsify H2, with one against 0 (Lakens 2017). *Choosing δ .* $\delta = 0.05$ is a provisional program decision threshold, not derived from theory. Before the confirmatory pre-registration it is justified, and signed off by the gate reader, against the cost of an expert hour, what a better-ranked area saves downstream, the cost of sending experts to areas that hold nothing, and the baselines’ AUROC in the pilot. *Planning reference.* Under independent errors and assumptions without empirical basis (Hanley–McNeil, AUROC 0.70 against 0.62, error correlation 0.5; Hanley and McNeil 1982), the 90% half-width of Δ AUROC is about ± 0.14 with 20 areas per class, ± 0.10 with 40, ± 0.06 with 100 and ± 0.05 with about 150; clustering widens all of these.

10.2 The studies

Each study below is given by its design, people, measure, sample size and gate; the full protocols and a per-study table are in the Extended Technical Report, Section 17.

Phase 0 — V1, the closure study (before any grant). **F** FUT. *Units:* a lens firing with its seed span, missing element and evidence pool, coded in both arms of the mismatched-evidence control. Of the 107 admissible firings (PLC 56, GDPR v1 51), 105 have a same-lens donor and enter both arms (210 pools); the two singletons are coded in their own arm only. A firing’s two arms are paired and firings share seed claims, sources and lenses, so the firing is the unit of inference; pools identical in both arms are coded once. About 100 pools from one new non-gold domain are added so that no judge is scored only in-sample. *Coders:* the judgment is textual, so coders need codebook training, not domain expertise. Preferred: two non-owner coders code every unit, blind to arm, judge answer, rank and run, and settle disagreements under the codebook; units still disputed are reported and left out of the criterion. The owner rules on codebook questions in a written log and codes no unit. NOW registers one non-owner coder, so this needs a recorded amendment; without it the owner is the second coder and MS1 is read as provisional until Phase 1 adds a second non-owner coder. *Retrieval recall:* for about 50 firings a coder searches the ledger’s full snapshots for the missing element. *Judges:* the local 7B model, a stronger open-weight model, frontier models from other families, and Jev, the decision model of the exploratory spike (Section 6.2). *Precision:* intervals resample seed claims with all their firings and both arms. Counted as 210 independent pools, $\kappa \approx 0.70$ would have a 95% half-width of about ± 0.10 ; counted as 107 firings, about ± 0.14 ; shared seeds and sources widen both, so these are floors. *Cost:* tens of coder hours per coder plus local compute, on the order of EUR 10^3 . *On STOP:* pivot to reconstruction as a documentation-audit tool, or to interviewing without a gap map.

Phase 1 — V0, V2 and V4. **F** FUT. V0 makes pipeline outputs admissible as code, builds the dated-corpus mode and the area score, and runs N2’s deferred protocols after a calibration pilot; its gate is the MS2 freeze (no hash, no gold). V2 scores the frozen gap map against several published CTA task lists, pooled. The lenses were built from CTA constructs, so V2 is a construct-aligned retrospective screen: a positive result shows the map is compatible with CTA-derived representations, not independent support for H2. A gold is eligible if it is an itemized list of steps, decisions or cues from more than one expert, in a domain other than PLC, GDPR or Track A physics, with a dated corpus above a floor. Named candidates, in order: Sullivan’s cricothyrotomy list, the only one confirmed as itemized (Sullivan et al. 2014); the NICU cue study (Crandall and Getchell-Reiter 1993); the troubleshooting study (Chao and Salvendy 1994); and the PARI guide (Hall et al. 1995). Proposed: at least three golds, four to six if eligible. Each gold domain is reconstructed once, from its dated corpus, before the sealed gold is acquired. Baselines include random, area size, retrieval density, corpus frequency, a type prior from the other golds and a plain-LLM ranking, and a text-form baseline (length statistics of the record text), since the spike showed such statistics can mimic a model’s score. *What MS3 can read:* under the independence planning figures, at four golds and 80 areas split evenly the half-width is about ± 0.10 , so CONTINUE needs a point estimate of at least 0.075; a map with no true gain passes with probability about 0.11, a true gain of δ with about 0.34. Areas nest in golds and three to six golds cannot estimate between-gold variance, so AUROC pairs areas only within a gold and these figures are optimistic. **I** INT. MS3 is a screen: it accepts about a one-in-nine chance of releasing Phase 2 for a useless map, which V5 then stops. V4 checks unseen-item estimators on V2 data against known truth; out of tolerance, only observed counts are reported. It does not gate.

Phase 2 — V5, blind prospective elicitation, the prospective test of H2. **F** FUT. *Design:* freeze reconstruction, dated corpus, gap map, area score and partition. In each domain, probe every area of the frozen partition, or a simple random sample of it if the partition exceeds the funded size. No predictor score decides which areas are probed, so ordinary AUROC estimates each predictor's AUROC over the partition. Each area gets a neutral probe of about 8 minutes, loosely adapted from the first pass of the CDM (Klein et al. 1989), a shortened format the CDM work does not test; it is the same for every area and never the gap map's own question. *Domains and people:* a PLC pilot, not analyzed for effect; the main study in two held-out domains. Practitioners with several years of practice, outside any Track A pool, three per area (Chao and Salvendy 1994); one interviewer per domain, two blind coders. *Outcome:* an item counts if it is absent from the frozen reconstruction *and* the dated corpus, and is corroborated by a second expert, a trace or an outside artifact; importance is rated blind to every score. The label is the residual *elicited* by this protocol: a short probe that finds nothing does not show that nothing is there (Section 9). *Blinding:* experts are told the study maps where practice relies on unwritten knowledge, not that a model scored areas (partial disclosure, stated in the ethics application, with debriefing). Interviewers see only area names in random order and no scores. *Sample size:* as a first-order reference only, independent Hanley–McNeil errors put the 90% half-width near δ at about 150 residual-present and 150 other areas; at that width MS5 stops a map with no gain, and continues one with a gain of δ , each only about half the time. Judgments nest in experts, areas share experts and each domain has one interviewer, so the real need is larger. The confirmatory size is set at MS4 by simulating the design over a stated range around the pilot's values. At the reference size: 900 probes, about 45 experts at 20 areas each plus 6 in the pilot, about 150–250 expert hours and 1,200–1,900 coder hours, roughly EUR 40–160k before staff time. *Gates:* MS4 releases the main study only if the pilot (6 experts $\times$ 20 areas, 40 areas) yields a simulated size of at most 450 areas that the partitions can supply, and segmentation reaches $\alpha \geq 0.80$; otherwise V5 stops and H2 is reported as untested prospectively. MS5 then reads its rule in the gate box.

Phase 2 — V6, V7, V8, only on MS5 CONTINUE. **F** FUT. V6 (H3, gate MS6) is a within-expert crossover of targeted questions against a standard unprompted-then-CDM interview; the breadth ranking and selection by expected information gain (not built) are separate arms (Lindley 1956). Items first seeded by a question count only if independently corroborated (Loftus 1975). CONTINUE if the lower 90% bound of the items-per-hour ratio exceeds 1 and the point estimate is at least 1.25; STOP if the upper bound is below 1.25; any other reading counts against H3. V7 (gate MS7) is a descriptive pilot in one partner organization with 6–12 staff experts, after the deferred N2 validation; it goes multi-partner only if G areas beat D and R in both parts. V8 covers the learner side: public learner data (no gate), 30–60 novices on confirmed residual areas (MS8), and Track A, touched only through sealed, secondary, non-gating predictions.

Organizational track — V3 (H-OSS). **F** FUT. None of its components exists: a Git adapter that turns commits, issues and reviews into claims with exact spans, areas mapped to code units at a time before a departure, and lenses for code text, frozen at MS2b once the adapter exists. The outcome is blind-coded rationale-seeking issues and defect-fix commits after truck-factor departures (Avelino et al. 2016), analyzed as a staggered difference-in-differences (Callaway and Sant'Anna 2021), against baselines that include authorship concentration and knowledge at risk (Rigby et al. 2016). A 5-departure pilot sizes the study; its gate uses the MS5 rule on H-OSS only. It can neither stop nor license H2.

I INT. Each STOP teaches something. At MS1 it shows that absence in text cannot yet be judged reliably, and it leaves a public closure set. At MS3 it leaves a reconstruct-versus-CTA benchmark.

At MS5 it bounds, in two domains, how well text reconstruction locates the residual that a short neutral probe elicits. Program threats (incomplete gold, method-dependent elicitation, matcher error) are treated there too.

11 Platform and applications

F **FUT.** The platform described here does not exist. The PoC is one walking skeleton toward it, and a gap candidate remains a prediction until a human answers it.

Three evidence worlds. Every piece of evidence belongs to one world, already a typed field in the code. *World A*, public evidence (textbooks, papers, standards, forums, public learner data), can yield concepts and taught heuristics, but not episode-specific cues or local practice. **O** **OBS.** It is implemented for open-web text only. *World B*, organization-specific evidence (procedures, tickets, commits, reviews, logs, incident reports), adds a second axis, practice: prescribed (imagined), recorded (done) or reported. **H** **HYP.** The gap between imagined and done is the organizational form of the residual. **O** **OBS.** The schema exists; no ingestion does. *World C*, human evidence (elicitation, ratings, learner responses, coder labels), has types but no data. **F** **FUT.** A and B build the reconstruction; the gap map predicts where C is needed; C answers through the channel matched to the predicted knowledge type and is the only path that can score the predictions. C never edits the reconstruction in place, and at least one human arm stays blind to the reconstruction, or anchoring shrinks the measured residual.

Rules the platform keeps. **O** **OBS.** Labels are computed, never set. A label is only as strong as its weakest link. Derived outputs (gap records, scores, questions) are inferred at best, and the evidential gate refuses them as a criterion. **F** **FUT.** Labels move up only through World C or cross-family verification; World B claims never leave their organization's tenant; and **unvalidated candidates never reach a learner or an end user.** No code enforces these three yet. On premises, the platform needs three model families available locally (generator, verifier, closure judge). **O** **OBS.** The PoC already runs its judge locally, which shows a local judge can be run, and also its risk: that 7B judge did not beat its mismatched-evidence control. Organizational deployment adds per-user access control propagated to claims, retrieved text treated as data and never as instructions, and a hash-chained audit trail (Extended Technical Report, Section 18).

O **OBS.** What exists today is the evidence ledger with its frozen schema, provenance verification on open web text, the gap map (candidates only), gates and checks, and a per-run call and cost log. Contradiction analysis beyond the older pairwise stage, company retrieval, expert elicitation for new tracks, residual measurement and expected-information-gain question selection are the next research stage; learner diagnostics, instructional design and adaptive learning are long-term. *Weft*, a separate public retrieval engine, was assessed as a possible future World B layer; it keeps no in-document character offsets and no per-user access control, so integration is not justified now, and at most it could supply candidate passages that the platform re-hashes and re-locates itself.

11.1 Education

L **LIT.** Expert judgment of learner difficulty is unreliable (Section 2), so the education path routes “what is difficult” through learner data alone. CTA-based instruction outperforms other ways of

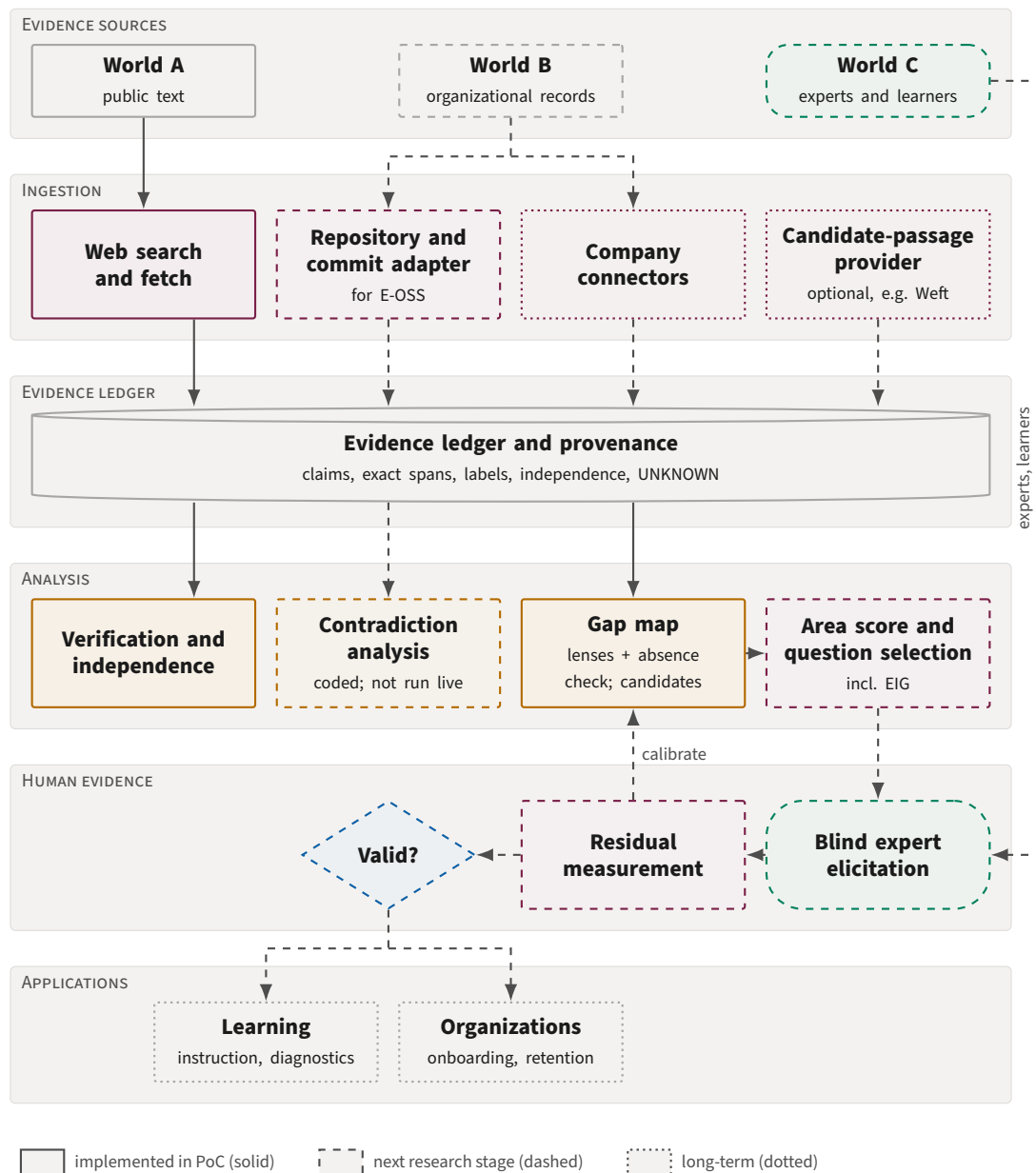


Figure 8: What would the full platform contain, and what exists today? Layers run from evidence sources through acquisition, provenance, the evidence ledger, analysis, the human loop and applications, with validation, audit and security as a rail. Solid boxes exist in the PoC; dashed boxes are the next research stage; dotted boxes are long-term. Only World C evidence can score the gap map's predictions.

identifying training content in a meta-analysis of a small number of studies ($g = 0.871$; Tofel-Grehl and Feldon 2013); in one biology course ($N = 314$) CTA-derived instruction cut withdrawal from 8.1% to 1.4% (Feldon et al. 2010). A worked example that prompts the learner to generate a step beats one that states it ($g = 0.35$, $k = 6$; Bisra et al. 2018). But a single item response is a poor diagnostic unit: 31% of Force Concept Inventory answers changed on a one-week retest (Lasry et al. 2011). A synthetic learner returns the misconception it was handed: a GPT-4 simulation of that inventory showed no response variance until told which preconception to hold (Kieser et al. 2023). And the median effect in 747 randomized education trials was 0.10 SD (Kraft 2020).

I INT. The pipeline is therefore evidence-complete only as far as “instructional material”, and only for items validated by humans. Synthetic learners serve only to pilot a probe or debug a pipeline, never as evidence. A candidate education pilot would add one or two expert-confirmed operations to self-explanation prompts in one course section and measure delayed transfer, as a feasibility pilot, not a powered test. Validation Track A (twelve physicists; about 370 students in a two-arm trial with delayed transfer, futility test at $d = 0.30$) runs only on its own preconditions, with its registered rules unchanged; this program touches it only through sealed, secondary, non-gating predictions.

11.2 Organizations

L LIT. Three organizational blind spots map onto constructs with literature behind them (the mapping is ours): a technician’s diagnostic cue, carried in stories rather than manuals (Orr 1996); a senior engineer’s design rationale, hard to transfer when cause and effect are ambiguous (Szulanski 1996); and a compliance officer’s exception judgment, the gap between work as imagined and work as done (Hollnagel 2018). Turnover loss is measurable in software: 65% of 133 popular GitHub projects have a truck factor of 2 or fewer (Avelino et al. 2016), and 315 of 1,932 projects (16%) were abandoned after truck-factor developers left (Avelino et al. 2019). **O OBS.** The PoC produced candidates of the first and third shape (the PLC GUARD record; a GDPR v2 record built on the “in most cases / in some cases” hedge of the DPIA guidance), both unvalidated, the second from an inadmissible run.

H HYP. Every value proposition is untested: that a gap map shortens onboarding, turns into a capture priority for a retiring expert, or adds predictive value beyond authorship concentration (which H-OSS tests). We found no credible, peer-reviewed cost figure for organizational knowledge loss in general and claim none. **I INT.** The limits are hard ones. A World B run needs access-scoped retrieval, on-premises or contractually isolated models, re-tuned lexicons, and a data protection impact assessment; where a German entity is involved, a works agreement. Outputs are aggregate by default, never feed performance evaluation, and name a person only with consent. A coverage score used as a target would reward documentation that raises the score over documentation that transfers knowledge. The recommended first partner is a software organization, closest to the registered H-OSS design.

12 Plan: domains, money, team and risks

F FUT. Everything in this section is future work, and every figure is a planning range.

12.1 Domains

A domain is chosen for the generalization question it answers and the human criterion that can test it, never for convenient data (Table 9). Three binding rules apply. Physics is the frozen Track A domain and enters only through sealed predictions. A held-out gold domain is never reconstructed before its configuration is frozen: its first reconstruction *is* the test run. PLC and GDPR tuned the lexicons, so they never carry a gate reading. Hence the recommended design: a V5 pilot in PLC, not analyzed for effect, and the V5 main study in a fresh maintenance-diagnosis task (D3) and a held-out regulation (D4).

Table 9: The six proposed domains. D1, D3 and D4 carry H2's gates; D2 carries H-OSS only; D5 and D6 gate nothing for H2.

Domain	Question	Criterion	Role
D1 Clinical procedure	Does the map find omitted steps and cues in a decision-heavy procedure?	Pooled published CTA goods, Sullivan's 46-item list first	V2; MS3 screen
D2 Open-source maintenance	Does artifact coverage predict realized knowledge loss?	Blind-coded trouble after truck-factor departures	V3 (H-OSS); V7 precursor
D3 Industrial fault diagnosis	Does it hold where craft dominates and text is poor?	Prospective blind elicitation	V5; MS5
D4 Regulatory judgment	Does it work where the residual sits in hedges and exceptions?	Prospective blind elicitation	V5; MS5
D5 School mathematics	Does an expert-text residual relate to learner difficulty?	Public learner response data	Learner link only
D6 Introductory mechanics	Best-case ceiling where residual, learner error and an RCT meet	Track A's own registered rules	Secondary, non-gating

12.2 Roadmap and the money each gate releases

Figure 9 shows the phases. Phase 0 runs alone and needs no grant, no participants and no new software. Phase 1 builds nothing for Phase 2, so no Phase 2 spending runs ahead of MS3. Table 10 gives each tranche. Personnel is costed at EUR 6–10k per person-month, to be replaced by the host institution's rate.

Table 10: Tranches, the gate that releases each, and indicative cost. Months count from the start of Phase 1.

Tranche	Released by	Scope	Months	PM	Total (EUR)
Phase 0	Owner, no grant	V1 closure study	2–4 mo before M1	–	≈ 10 ³
Phase 1	MS1 continue	Area score, freeze, V0, judge check, pooled V2, V4	M1–M14	35–44	0.23–0.51 M
2a	MS3 continue	Workbench, V5 pilot, learner data, ledger groundwork	M15–M24	29–44	0.19–0.49 M
2b	MS4 continue	V5 main, sized at MS4	M25–M38	22–36	0.17–0.61 M
2c	MS5 continue	V6, V7, novice study, education pilot, on-prem stack	M39–M48	38–62	0.26–0.79 M
O	MS3 continue	Git adapter, code lenses, V3 (H-OSS)	M15–M36	14–22	0.09–0.26 M
Phase 2 Program		2a–2c and O	M15–M48	103–164	0.7–2.2 M
		Phases 1 and 2	48 mo	138–208	0.9–2.7 M
Interview-only pivot	MS1, MS3 or MS5 stop	Separate proposal	18–24 mo	21–31	0.14–0.38 M

1 INT. Money at risk if H2 fails: at MS1, about EUR 10³ plus the owner's time; at MS3, Phase 1 (EUR 0.23–0.51 M), which still leaves a released closure set, a frozen area score and a public benchmark; at MS4, EUR 0.42–1.00 M; at MS5, EUR 0.59–1.61 M, and tranche 2c is never released. The binding constraint of V5 is recruiting about 45 eligible experts, not money. **o OBS.** The PoC's one hard cost number: a reconstruction run cost \$0.50–\$1.00. Money was not the constraint; headroom was. **1 INT.** So provider limits are set at least three times the sum of a stage's registered caps, the headroom check runs as code, each experiment has its own key, and every study reports cost per *validated* item.

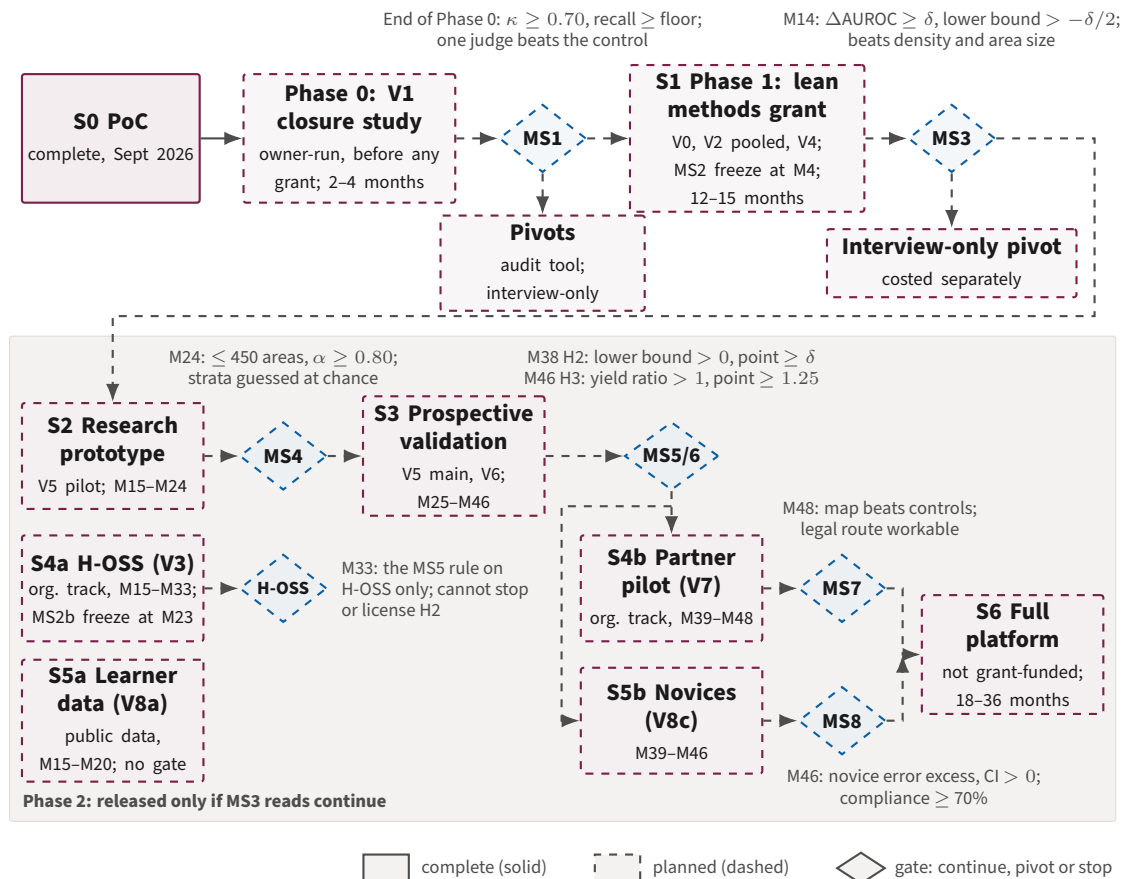


Figure 9: Which gated stages lead from PoC to platform? Phase 0 (owner-run, before any grant) is the closure study, ending at MS1. Phase 1, a lean methods grant, starts only on MS1 CONTINUE and ends at the screening gate MS3. Phase 2 starts only on MS3 CONTINUE; its prospective study tests H2 at MS5. Every stop leads to a pivot with its own, smaller budget. Only S0, the PoC, is complete.

12.3 Team

F FUT. FUTURE WORK

Team status. No team exists yet: no host institution, no principal investigator (PI) record for this program, no statistician, no CTA collaborator, no gate reader and no letter of support. One operator designed, built, ran and read the PoC with AI assistance, and is the prospective applicant. Before MS1 is read: the non-owner blind coders and the external gate reader. Before the Phase 1 proposal: host institution, PI, statistician as co-applicant, CTA collaborator, research engineer, data-protection and license advisor. Before Phase 2: a cognitive psychologist or CTA lead, expert pools with letters from professional bodies, and a learning scientist.

I INT. The hard problems span language processing and measurement, CTA, learning science, statistics, and security, privacy and law, so the team is interdisciplinary. It grows only when a gate releases the next phase (Table 11).

I INT. One staffing rule cannot be relaxed by adding people: an advisor who saw a domain's gap map may not later be interviewed, code, rate or corroborate in that domain; the pipeline's

Table 11: Team by phase. Phase 1 (12–15 months, about 2.5–3.0 FTE, 35–44 person-months) reaches MS3; Phase 2 (34 months, about 3.0–4.8 FTE on average across 6–8 people) is hired only on MS3 CONTINUE. Full role tables: Extended Technical Report, Section 24 and Appendix H.

Role	Phase 1 FTE	Phase 2 FTE	Main work
PI / research lead	0.8–1.0	0.5–1.0	Hypotheses, thresholds, gate reports (never reads a gate)
Research engineer (LLM / retrieval)	1.0	0.5–1.0	Area score, freeze, dated corpora; judge upkeep, on-prem models
Research assistant or second engineer	0.5	–	Gold availability, partitions, memorization probes
Statistician / methodologist	0.3	0.3–0.5	Threshold table, pooled and clustered AUROC
CTA practitioner / cognitive psychologist	0.1–0.2	0.5 → 1.0	Codebook, gold mapping; V5 interviewing
AI/ML researcher	–	1.0	Prospective designs, baselines, repository mining
Platform engineer	–	1.0	Git adapter, workbench back end, audit trail
Learning scientist; HCI; project manager; legal	–	part-time	Learner link; leak-proof interfaces; partner contracts; DPIA
Coders	160–380 h	1,000–2,000 h	Blind double coding
External gate reader	per gate	per gate	Reads MS1, MS3, MS5; independent

owner is never a primary coder on a gating score where a second non-owner coder can be found; interviewers and coders never see scores.

12.4 Top risks and kill criteria

Table 12: Risks that can stop or redirect the program (likelihood / impact, qualitative). Full table: Extended Technical Report, Section 26.

Risk	L/I	Early signal	Kill / pivot criterion
Absence check cannot be made informative	H/H	 OBS. Own rate equals control rate on every ledger	MS1 fails → stop the automated absence check; no grant for the gap map
Gap map does not beat the strongest baseline	M/H	PLC ranking tracks density ($\rho = 0.91$)	MS3 fails → no Phase 2 money; MS5 short of CONTINUE → H2 not supported
Too few golds	M/H	Fewer than three eligible itemized golds	MS3 rule unchanged, so CONTINUE gets harder; nothing stands in
Contamination	H/M	Many probe-positive items	Drop that gold from the pooled reading
Expert recruitment	M/H	Pilot projects more than 450 areas	MS4 → V5 main not released; H2 untested
Owner overrides a gate	M/H	An exception like N2 → N3	Without reader sign-off the reading is void
Team / key person	H/H	No host, no named team	Program pauses at its last gate
Budget headroom	L– M/M	Limit below the sum of caps	Rerun with headroom; never read as negative
Ethics / legal (V7)	M/H	DPIA risk or works-council objection	Stop the pilot at that partner

 **INT.** Two pivots exist, each on its own budget. The *interview-only pivot* compares question selection by expected information gain with untargeted interviewing, without any reconstruction prior; a negative result bounds how far text reconstruction can go. The *documentation-audit pivot* keeps the ledger as a provenance-checked map of what a text states, with no claim about hidden

knowledge.

12.5 Expected contributions

OBS. Three contributions exist today, independent of H2: a provenance-typed evidence model, self-checks built into every gap map, and documented negative and inconclusive results. The novelty ledger grades them as known methods, known combinations or engineering. **FUT.** Table 13 lists what the program would add; every row delivers something even on a stop.

Table 13: Expected contributions, graded by outcome. All rows are future work.

Contribution (study)	If it holds	Even if it fails
Human-labeled closure set (V1)	HYP. A reproducible human label for “not stated in this evidence”, and a test of whether any judge tracks it	A measured answer to whether absence can be judged; a public benchmark
Pooled CTA-gold benchmark (V2)	HYP. A first retrospective Δ AUROC, conditional on density	A contamination-controlled test for any system that claims to find what text omits
Post-departure dataset (V3)	HYP. Coverage predicts loss beyond authorship	Evidence that it does not; a public outcome set
Residual corpus (V5)	HYP. Calibration data for gap maps	A typed record of what experts add beyond a fixed text
Efficiency (V6)	HYP. At least 1.25 times more validated items per expert hour	A bound on the gain
Learner link (V8)	HYP. Residual areas carry more novice errors	An upper bound on how far residual areas track novice errors in this sample

INT. A pre-registered negative in V5 would itself be a contribution: a bounded falsification, of a kind we found nowhere published.

13 Conclusion

OBS. The chain exists and runs: computed labels, gates that refuse synthetic criteria, evidence items that re-slice exactly from hashed snapshots, and a gap map that replays byte for byte. This supports H1 as an engineering result; its adversarial part is open, since N2’s registered verdict is INCONCLUSIVE. The absence check is the crux, and it does not work yet: its closure rate was the same with a candidate’s own additional evidence and with its nearest neighbor’s (0.889, 0.961, 0.958), while both arms kept the seed’s sentences. **INT.** The PoC neither demonstrates nor refutes H2. An exploratory comparator, Jev, gave no interpretable hidden-knowledge signal.

FUT. Next action: run the closure study (V1) before any grant submission, on the 107 admissible firings (105 in both control arms) plus about 100 units from a new domain, coded blind by non-owner coders. It needs no participants and costs on the order of EUR 10³. Only on MS1 CONTINUE does a lean methods grant follow.

FINDING

AI can already build a record, checkable span by span, of claims drawn from a sample of written sources about a task; its accuracy and completeness have not been measured. Whether its silences mark what experts know is untested. That test starts with the absence check, and the absence check can be tested now, before any grant.

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